~~Consensus Control of Switched Multi-Agent Systems with State-Dependent Dwell-Time Switching via Lyapunov–Metzler Inequalities~~

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***Abstract*—~~This paper investigates the consensus problem for switched multi-agent systems with multiple candidate directed communication topologies, where every candidate topology is disconnected while the union graph of all candidate topologies is strongly connected. First, a consensus-orthogonal decomposition is introduced to transform the original multi-agent system into a reduced-order disagreement error system. Then, the strong connectivity of the union graph is used to rule out common non-consensus zero directions, thereby showing that the system has no structural obstruction to achieving consensus through switching. Based on this reduced-order formulation, a state-dependent dwell-time switching law designed using Lyapunov–Metzler inequalities stabilizes the disagreement error system. Rather than stabilizing the system through a preassigned Hurwitz average topology, the proposed method directly designs a state-dependent dwell-time switching law based on Lyapunov–Metzler inequalities. Instead, the cooperative stabilizing property among multiple disconnected topologies is characterized through Lyapunov–Metzler inequalities and realized by the proposed switching law. Furthermore, the proposed switching mechanism guarantees a prescribed dwell-time constraint, thereby avoiding arbitrarily fast switching. It is shown that, under the feasibility of the Lyapunov–Metzler inequalities, the reduced-order disagreement error converges to zero, and hence global consensus of the original multi-agent system is achieved. Finally, a numerical example with two individually disconnected candidate topologies is provided to verify the effectiveness of the theoretical results.~~**

***Index Terms*—Multi-agent ~~system, consensus~~, ~~switched system, strong union graph connectivity~~, Lyapunov–Metzler ~~inequality~~, state-dependent dwell-time ~~switching~~.**

I. Introduction

Consensus control is a fundamental problem in multi-agent systems, where a group of agents is required to reach an agreement ~~by using only~~ local information exchange over ~~a~~ communication network [1], [2]. Owing to its broad applications in cooperative control, distributed sensing, formation control, unmanned systems, and networked autonomous systems, consensus has been extensively studied ~~under both fixed and switching communication topologies~~. For directed networks, a typical graph-theoretic condition for consensus is that the communication graph is strongly connected or contains a directed spanning tree [3], [4]. ~~Under such conditions, the Laplacian matrix has a simple zero eigenvalue associated with the consensus direction, and the disagreement dynamics can be analyzed after removing this direction.~~

In practical networked systems, however, ~~the available communication topology may be incomplete~~ due to link failures, sensing limitations, communication constraints, cyber attacks, or intermittent information exchange. ~~As a result, an individual topology may not be strongly connected~~ and may fail to guarantee global consensus by itself. Nevertheless, ~~several incomplete topologies may jointly provide sufficient information paths.~~ This motivates the study of ~~switching topologies under joint or union graph connectivity.~~ Existing results have shown that consensus can still be achieved when the switching graphs satisfy jointly connected conditions, even though each instantaneous graph may not be connected [5]–[7].

~~A standard approach for consensus analysis is to separate the consensus component from the disagreement component.~~ By using an orthogonal or linear transformation, the original consensus problem can be converted into the stability problem of a reduced-order disagreement error system. This idea has been widely used in multi-agent consensus analysis. For instance, orthogonal transformation techniques have been used to characterize consensus performance and disturbance attenuation, while linear transformations have been adopted to convert delayed consensus problems into robust stability problems associated with Laplacian eigenvalues [8], [9]. In this paper, a consensus-orthogonal decomposition is introduced to remove the consensus direction and obtain a redundancy-free switched disagreement error system.

~~The problem considered in this paper is also related to the stability of switched systems whose individual subsystems are not necessarily stable.~~ Classical switched-system theory has shown that a switched system can be stabilized by a suitable switching law even when not all subsystems are stable [10], [11]. For switched systems with stable and unstable subsystems, average dwell-time and multiple Lyapunov function techniques have been extensively developed [13]. Recent studies have further investigated switched systems with unstable modes, including mode-partition-dependent average dwell-time methods [15], mixed stable–unstable subsystems with multiple equilibria [16], all-mode-unstable switching systems [17], and discrete-time switched systems with all unstable subsystems [18]. These studies indicate that instability of individual subsystems does not necessarily prevent stability of the overall switched system.

Recently, Russo et al. proposed a state-dependent dwell-time switching framework for switched affine systems based on differential Lyapunov equations and Lyapunov–Metzler inequalities [19]. Their method shows that stabilization can be achieved without relying on the existence of a Hurwitz [illegible]k combination of subsystem matrices. Inspired by this [illegible] this paper applies a Lyapunov–Metzler-based dwell-[illegible]witching mechanism to the reduced-order disagreement [illegible]ics of switched multi-agent systems, so that a prescribed minimum dwell time can be enforced while the active topology is selected according to the current disagreement state.

Motivated by the above observations, this paper studies the consensus problem for switched multi-agent systems with multiple candidate directed communication topologies. Different from conventional consensus results that require each individual topology to be strongly connected or to contain a directed spanning tree, this paper considers the case where each candidate topology is not strongly connected, while the union graph of all candidate topologies is strongly connected. The main idea is to use the union graph connectivity to exclude common non-consensus zero directions and then design a state-dependent dwell-time switching law via Lyapunov–Metzler inequalities to stabilize the reduced-order disagreement system.

The main contributions of this paper are summarized as follows:

- A consensus-orthogonal decomposition is employed to transform the original multi-agent consensus problem into a reduced-order switched disagreement error system.
- The strong connectivity of the union graph is used to show that there is no common non-consensus zero direction among the candidate topologies, which provides a structural basis for consensus under switching.
- A Lyapunov–Metzler inequality condition is established for the reduced-order error system without requiring any individual topology subsystem to be Hurwitz.

The rest of this paper is organized as follows. Section II presents the system model and derives the reduced-order disagreement error system. Section III analyzes the role of strong union graph connectivity in excluding common non-consensus zero directions. Section IV develops the Lyapunov–Metzler-based dwell-time switching law and proves the stability result. Section V provides numerical simulations to verify the proposed method. Finally, Section VI concludes the paper.

Notations: Let $\mathbb{R}^d$ denote the d-dimensional Euclidean space. $\mathbf{1}_N$ denotes the N-dimensional column vector of all ones, and I_N denotes the $N \times N$ identity matrix. For a matrix A, $A^\top$ denotes its transpose. For a symmetric matrix A, $A > 0$ (resp. $A < 0$) means that A is positive definite (resp. negative definite). The symbol $\otimes$ denotes the Kronecker product. The notation $\|\cdot\|$ denotes the Euclidean norm. We denote by $\mathcal{M}$ the class of irreducible Metzler matrices, which consists of all matrices $\Pi = [\pi_{ij}] \in \mathbb{R}^{M\times M}$ such that $\pi_{ij} \geq 0$ for all $i \neq j$ and $\sum_{j=1}^{M} \pi_{ij} = 0$ for all $i \in \{1, \ldots, M\}$.

II. Preliminaries

Consider a multi-agent system consisting of N agents, where the state of agent i is $x_i(t) \in \mathbb{R}^d$, $i = 1, 2, \ldots, N$. Define the stacked state vector as $x(t) = \left[x_1^\top(t), x_2^\top(t), \ldots, x_N^\top(t)\right]^\top \in \mathbb{R}^{Nd}$. Suppose that the communication network can switch among M candidate directed topologies $\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_M$~~, each associated with a row-sum-zero Laplacian matrix~~ $\cancel{L_p,\ p = 1, 2, \ldots, M}$.

For each directed topology $\mathcal{G}_p$, the adjacency weight a_{ij}^p represents the influence of agent j on agent i. The associated Laplacian matrix L_p is defined by $[L_p]_{ii} = \sum_{j\neq i} a_{ij}^p$ and $[L_p]_{ij} = -a_{ij}^p$ for $i \neq j$, which implies $L_p \mathbf{1}_N = 0$, $p = 1, 2, \ldots, M$.

The communication structure considered in this paper is specified by the following assumption.

Assumption 1. Each candidate directed topology $\mathcal{G}_p$, $p = 1, 2, \ldots, M$, is disconnected, while the union graph of all candidate directed topologies, denoted by $\mathcal{G}_U = \bigcup_{p=1}^{M} \mathcal{G}_p$, is strongly connected.

This assumption indicates that no individual candidate topology can guarantee global consensus by itself. Nevertheless, although each candidate topology only provides partial directed information-exchange paths, the collection of all candidate topologies covers the entire agent network in the sense of strong connectivity. Therefore, global consensus may be achieved through an appropriate switching strategy.

Under the $\sigma(t)$-th topology, the following standard consensus control protocol is adopted:

$$u_i(t) = -\sum_{j=1}^{N} a_{ij}^{\sigma(t)} \left(x_i(t) - x_j(t)\right), \quad i = 1, 2, \ldots, N, \tag{1}$$

where $a_{ij}^{\sigma(t)}$ denotes the directed information weight from agent j to agent i under the current topology $\mathcal{G}_{\sigma(t)}$, and $\sigma(t) \in \Omega = \{1, 2, \ldots, M\}$ is the switching signal.

Stacking all control inputs as $u(t) = \begin{bmatrix} u_1^\top(t) & u_2^\top(t) & \cdots & u_N^\top(t)\end{bmatrix}^\top$, the overall system can be written as

$$\dot{x}(t) = u(t) = -\left(L_{\sigma(t)} \otimes I_d\right) x(t), \tag{2}$$

where I_d is the d-dimensional identity matrix.

~~Define~~ $\cancel{A_p = -(L_p \otimes I_d),\ p = 1, 2, \ldots, M}$. ~~Then the system can be formally expressed as a switched linear system~~ $\cancel{\dot{x}(t) = A_{\sigma(t)} x(t)}$.

~~It should be noted that, although the above form resembles a general switched linear system, the objective of this paper is not to drive the original state~~ $\cancel{x(t)}$ ~~to the origin.~~ Instead, consensus is achieved when $\lim_{t\to\infty}(x_i(t) - x_j(t)) = 0$ for all $i, j \in \{1, 2, \ldots, N\}$, or equivalently when $x(t)$ converges to the consensus subspace $\mathcal{S}_c = \{\mathbf{1}_N \otimes \eta : \eta \in \mathbb{R}^d\}$. ~~Therefore,~~

~~origin stability analysis of $x(t)$ is not appropriate, and the consensus problem should be reformulated as the stability problem of a disagreement error system.~~

To describe the disagreement among agents, define the centering matrix $H = I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top$ and the average state $\bar{x}(t) = \frac{1}{N}(\mathbf{1}_N^\top \otimes I_d)x(t)$. The disagreement error is then given by $e(t) = (H \otimes I_d)x(t)$, and the state can be decomposed as $x(t) = e(t) + \mathbf{1}_N \otimes \bar{x}(t)$. Hence, $e(t)$ represents the deviation of agents from their common average, and $e(t) \to 0$ is equivalent to $x_i(t) - x_j(t) \to 0$ for all $i, j \in \{1, 2, \ldots, N\}$.

Since $L_{\sigma(t)}\mathbf{1}_N = 0$, the Laplacian term only acts on the disagreement component. Indeed, from $x(t) = e(t) + \mathbf{1}_N \otimes \bar{x}(t)$, one has

$$\begin{aligned}\left(L_{\sigma(t)} \otimes I_d\right) x(t) &= \left(L_{\sigma(t)} \otimes I_d\right) e(t) + \left(L_{\sigma(t)}\mathbf{1}_N\right) \otimes \bar{x}(t)\\ &= \left(L_{\sigma(t)} \otimes I_d\right) e(t).\end{aligned}$$

Thus, the consensus component $\mathbf{1}_N \otimes \bar{x}(t)$ is not affected by the Laplacian matrix, and only the disagreement error contributes to the system dynamics.

Although $e(t)$ directly represents the consensus error, it is a redundant variable since it always satisfies $(\mathbf{1}_N^\top \otimes I_d)e(t) = 0$. Hence, $e(t)$ evolves on a constrained consensus-orthogonal subspace of dimension $(N-1)d$, and a Lyapunov function based on $e(t)$ would require positive definiteness to be considered on this subspace.

To obtain a redundancy-free disagreement coordinate, choose a matrix $S \in \mathbb{R}^{(N-1)\times N}$ such that $S\mathbf{1}_N = 0$ and $\operatorname{rank}(S) = N - 1$. Thus, the rows of S span the subspace orthogonal to the consensus direction. Without loss of generality, S can be chosen to satisfy $SS^\top = I_{N-1}$ and $S^\top S = H$. Define the reduced-order disagreement error as $\xi(t) = (S \otimes I_d)x(t)$. Since $S\mathbf{1}_N = 0$, any consensus state $x(t) = \mathbf{1}_N \otimes \eta(t)$ satisfies $\xi(t) = 0$. Hence, $\xi(t) \to 0$ is equivalent to consensus of the multi-agent system.

From $\xi(t) = (S \otimes I_d)x(t)$ and $\dot{x}(t) = -(L_{\sigma(t)} \otimes I_d)x(t)$, one obtains

$$\dot{\xi}(t) = -\left(SL_{\sigma(t)} \otimes I_d\right) x(t).$$

Using the decomposition $x(t) = \mathbf{1}_N \otimes \bar{x}(t) + (S^\top \otimes I_d)\xi(t)$, it follows that

$$\dot{\xi}(t) = -\left(SL_{\sigma(t)}\mathbf{1}_N \otimes I_d\bar{x}(t)\right) - \left(SL_{\sigma(t)}S^\top \otimes I_d\right)\xi(t).$$

~~Since $L_{\sigma(t)}\mathbf{1}_N = 0$, hence,~~

$$\dot{\xi}(t) = -\left(SL_{\sigma(t)}S^\top \otimes I_d\right)\xi(t). \tag{3}$$

Define $\bar{A}_p = -\left(SL_pS^\top \otimes I_d\right), \quad p = 1, 2, \ldots, M$. The consensus error system can then be rewritten as the standard switched linear system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t). \tag{4}$$

Remark 1. The introduction of $\xi(t)$ is not intended to decouple the system into $N-1$ independent subsystems. Instead, it removes the consensus direction and provides a redundancy-free disagreement error state. Therefore, the subsequent stability analysis is carried out on the overall $(N-1)d$-dimensional switched error system $\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t)$.

III. Strong Union Graph Connectivity and Common Non-Consensus Zero Directions

Since each candidate directed topology $\mathcal{G}_i$ is not strongly connected, the corresponding reduced matrix $SL_iS^\top$ may be singular, and a single topology generally cannot eliminate all disagreement directions. However, the union graph $\mathcal{G}_U = \bigcup_{i=1}^{M}\mathcal{G}_i$ is assumed to be strongly connected. Let its Laplacian be

$$L_U = \sum_{i=1}^{M} L_i.$$

Then the row-sum-zero Laplacian L_U satisfies

$$\ker(L_U) = \operatorname{span}\{\mathbf{1}_N\}.$$

Since $S\mathbf{1}_N = 0$, the consensus direction is removed in the reduced-order error coordinates, and hence

$$\ker\left(SL_US^\top \otimes I_d\right) = \{0\}.$$

This implies that the candidate topologies have no common non-consensus zero direction in the disagreement error space.

Lemma 1. If the union graph $\mathcal{G}_U$ of all candidate directed topologies is strongly connected, then

$$\bigcap_{p=1}^{M} \ker\left(SL_iS^\top \otimes I_d\right) = \{0\}.$$

Proof. Suppose that there exists $\xi^* \neq 0$ such that

$$\left(SL_iS^\top \otimes I_d\right)\xi^* = 0, \quad i = 1, 2, \ldots, M.$$

Then, using $L_U = \sum_{i=1}^{M} L_i$, one obtains

$$\left(SL_US^\top \otimes I_d\right)\xi^* = 0.$$

Since the strong connectivity of $\mathcal{G}_U$ gives

$$\ker\left(SL_US^\top \otimes I_d\right) = \{0\},$$

it follows that $\xi^* = 0$, which contradicts $\xi^* \neq 0$. Hence, the conclusion holds. $\square$

Remark 2. The strong connectivity of the union graph guarantees that there is no common non-consensus zero direction in the system, i.e., it structurally excludes those error directions that cannot be eliminated by any switching strategy.

IV. Lyapunov–Metzler-Based Dwell-Time Switching and Stability Analysis

Consider the reduced-order disagreement error system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t), \quad \bar{A}_i = -\left(SL_iS^\top \otimes I_d\right), \quad i = 1, 2, \ldots, M. \tag{5}$$

To characterize the disagreement performance, define the output $z(t) = C\xi(t)$, where C is a given matrix. If the whole disagreement error is to be measured directly, one may take $C = I$.

For each $\mathcal{G}_i$, choose a positive definite matrix $X_i = X_i^\top > 0$ and define the Lyapunov function as

$$V_i(\xi) = \xi^\top X_i\xi. \tag{6}$$

Let $\Pi = [\pi_{ij}] \in \mathcal{M}$ be an irreducible Metzler matrix satisfying $\pi_{ij} \geq 0$ for $i \neq j$ and $\sum_{j=1}^{M} \pi_{ij} = 0$. Given a prescribed minimum dwell time $T > 0$, define, for each mode j,

$$Y_{1,j} = e^{\bar{A}_j^\top T} X_j e^{\bar{A}_j T}, \tag{7}$$

and

$$Y_{2,j} = \int_0^T e^{\bar{A}_j^\top \tau} C^\top C e^{\bar{A}_j \tau} d\tau. \tag{8}$$

Here, $Y_{1,j}$ denotes the predicted Lyapunov value after the system switches to j and remains there for the dwell time T, while $Y_{2,j}$ denotes the accumulated disagreement energy over this interval. Thus, $Y_{1,j} + Y_{2,j}$ can be interpreted as the predicted dwell-time cost associated with j.

The following Lyapunov–Metzler inequalities are used as the main stability condition:

$$\begin{aligned} &\bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j \neq i} \pi_{ij} \left(Y_{1,j} + Y_{2,j} - X_i \right) + C^\top C < 0, \\ &i = 1, 2, \ldots, M. \end{aligned} \tag{9}$$

These inequalities do not require any individual subsystem matrix $\bar{A}_i$ to be Hurwitz, nor do they require the existence of a Hurwitz average matrix. Instead, the Metzler coupling term is used to characterize the cooperative stabilizing effect among different topology.

For a given active mode $i = \sigma(t_k)$, define

$$\Omega_i = \{1, 2, \ldots, M\} \setminus \{i\}.$$

For $j \in \Omega_i$, let

$$\Delta_{ij}(t) = \xi^\top(t) \left(Y_{1,j} + Y_{2,j} - X_i \right) \xi(t),$$

and

$$J_j(t) = \xi^\top(t) \left(Y_{1,j} + Y_{2,j} \right) \xi(t).$$

Then, the proposed state-dependent dwell-time switching law is given by

$$\begin{aligned} &\sigma(t) = i, && t \in [t_k, t_k + T], \\ &\sigma(t) = i, && t > t_k + T,\ \Delta_{ij}(t) \geq 0,\ \forall j \in \Omega_i, \\ &\sigma(t_{k+1}) = \arg\min_{j \in \Omega_i} J_j(t_{k+1}), && \text{otherwise.} \end{aligned} \tag{10}$$

where

$$t_{k+1} = \inf_{t > t_k + T} \left\{ t : \exists j \in \Omega_i,\ \Delta_{ij}(t) < 0 \right\}. \tag{11}$$

Theorem 1. Consider the reduced-order disagreement error system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)} \xi(t).$$

Assume that Assumption 1 holds. If there exist positive definite matrices $X_i = X_i^\top > 0$, $i = 1, 2, \ldots, M$, and an irreducible Metzler matrix Π such that the Lyapunov–Metzler inequalities

$$\bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j \neq i} \pi_{ij} \left(Y_{1,j} + Y_{2,j} - X_i \right) + C^\top C < 0 \tag{12}$$

hold for all $i = 1, 2, \ldots, M$, then under the proposed state-dependent dwell-time switching law, the disagreement error system is asymptotically stable. In particular,

$$\xi(t) \to 0, \quad t \to \infty.$$

Consequently, the original multi-agent system achieves global consensus, i.e.,

$$\lim_{t \to \infty} \left(x_i(t) - x_j(t) \right) = 0, \quad \forall i, j.$$

Proof. Let t_k be a switching instant and suppose that the active mode is i. Define the following piecewise Lyapunov function:

$$V(\xi, t) = \xi^\top(t) P_i(t) \xi(t), \quad t \in [t_k, t_k + T), \tag{13}$$

$$V(\xi, t) = \xi^\top(t) X_i \xi(t), \quad t \in [t_k + T, t_{k+1}). \tag{14}$$

Here, $P_i(t)$ satisfies

$$-\dot{P}_i(t) = \bar{A}_i^\top P_i(t) + P_i(t) \bar{A}_i + C^\top C, \tag{15}$$

with the terminal condition

$$P_i(t_k + T) = X_i. \tag{16}$$

On the forced dwell-time interval $[t_k, t_k + T)$, one has $\sigma(t) = i$. Using the error dynamics and the above differential Lyapunov equation gives

$$\begin{aligned} \dot{V}(\xi, t) &= \xi^\top(t) \left(\bar{A}_i^\top P_i(t) + P_i(t) \bar{A}_i + \dot{P}_i(t) \right) \xi(t) \\ &= -\xi^\top(t) C^\top C \xi(t). \end{aligned} \tag{17}$$

Thus, the Lyapunov function is non-increasing during the forced dwell-time interval.

For $t \in [t_k + T, t_{k+1})$, no switch has been triggered. Hence,

$$\xi^\top(t) \left(Y_{1,j} + Y_{2,j} - X_i \right) \xi(t) \geq 0, \quad \forall j \neq i.$$

Since $\pi_{ij} \geq 0$, it follows from the Lyapunov–Metzler inequality that

$$\begin{aligned} \dot{V}(\xi, t) &= \xi^\top(t) \left(\bar{A}_i^\top X_i + X_i \bar{A}_i \right) \xi(t) \\ &< -\xi^\top(t) \left[\sum_{j \neq i} \pi_{ij} \left(Y_{1,j} + Y_{2,j} - X_i \right) + C^\top C \right] \xi(t) \\ &= -\xi^\top(t) \left[\sum_{j \neq i} \pi_{ij} \left(Y_{1,j} + Y_{2,j} - X_i \right) \right] \xi(t) \\ &\quad - \xi^\top(t) C^\top C \xi(t) \\ &\leq -\xi^\top(t) C^\top C \xi(t). \end{aligned} \tag{18}$$

Therefore, the Lyapunov function remains non-increasing before the next switching instant.

At the switching instant t_{k+1}, the switching condition ensures that there exists some $j \neq i$ satisfying

$$\xi^\top(t_{k+1}) \left(Y_{1,j} + Y_{2,j} - X_i \right) \xi(t_{k+1}) < 0. \tag{19}$$

Moreover, the new mode is selected by minimizing the predicted dwell-time cost. Therefore,

$$\begin{aligned}\Delta V(t_{k+1}) &= V(\xi(t_{k+1}), t_{k+1}) - V(\xi(t_{k+1}^-), t_{k+1}^-) \\ &\le \xi^\top(t_{k+1})\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t_{k+1}) \\ &< 0.\end{aligned} \tag{20}$$

Thus, each switching instant yields a strict decrease of the Lyapunov function.

Combining the continuous evolution and the switching instants, one obtains

$$\dot{V}(\xi, t) \le -\xi^\top(t) C^\top C \xi(t), \tag{21}$$

$$\Delta V(t_k) < 0. \tag{22}$$

Consequently, for any $t \ge t_0$,

$$\int_{t_0}^{t} \xi^\top(\tau) C^\top C \xi(\tau) d\tau \le V(\xi(t_0), t_0). \tag{23}$$

Together with the positive definiteness of the Lyapunov function and standard stability arguments, this implies $\xi(t) \to 0$. Since $\xi(t) \to 0$ is equivalent to $x_i(t) - x_j(t) \to 0$ for all i, j, the original multi-agent system achieves global consensus. □

V. Numerical Simulations

To validate the proposed consensus control method based on Lyapunov–Metzler inequalities and the state-dependent dwell-time switching law, this section presents a numerical example with two disconnected candidate topologies. This example demonstrates that, although each candidate topology is individually not strongly connected and cannot achieve global consensus alone, the system can achieve global consensus under the strong union graph connectivity condition through the designed state-dependent dwell-time switching law.

A. Simulation Setup

Consider a first-order multi-agent system consisting of $N = 3$ agents with $d = 1$, i.e., each agent's state is $x_i(t) \in \mathbb{R}$. The system has $M = 2$ candidate directed communication topologies. The corresponding Laplacian matrices are given by

$$L_1 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad L_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Topology $\mathcal{G}_1$ only connects agents 1 and 2, while topology $\mathcal{G}_2$ only connects agents 2 and 3. Hence, neither topology is strongly connected, and each topology alone can only achieve local consensus among part of the agents.

The union graph is $\mathcal{G}_U = \mathcal{G}_1 \cup \mathcal{G}_2$, whose Laplacian matrix is

$$L_U = L_1 + L_2 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

The eigenvalues of L_U are $\{0, 1, 3\}$, which implies that the union graph is strongly connected and that the two candidate topologies have no common non-consensus zero direction.

B. Reduced-Order Error System

Choose the consensus-orthogonal matrix

$$S = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \end{bmatrix},$$

which satisfies $S\mathbf{1}_3 = 0$ and $SS^\top = I_2$. The reduced-order disagreement error is defined as $\xi(t) = Sx(t)$, and the corresponding error dynamics are

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t), \quad \bar{A}_i = -SL_iS^\top, \quad i = 1, 2.$$

Direct calculation gives

$$\bar{A}_1 = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}, \quad \bar{A}_2 = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{3}{2} \end{bmatrix}.$$

Both matrices have eigenvalues $\{-2, 0\}$, implying that neither reduced-order subsystem is Hurwitz. This is consistent with the fact that each candidate topology alone cannot achieve global consensus. In contrast, the reduced union-graph matrix $SL_US^\top$ has eigenvalues $\{1, 3\}$ and hence $SL_US^\top > 0$, which confirms that the two candidate topologies jointly cover all disagreement directions.

C. Lyapunov–Metzler Parameter Configuration

Set $C = I_2$ and choose the minimum dwell time as $T = 0.2$. The Lyapunov matrices and the Metzler matrix are selected as

$$X_1 = 444I_2, \quad X_2 = 519I_2, \quad \Pi = \begin{bmatrix} -0.27 & 0.27 \\ 0.27 & -0.27 \end{bmatrix}.$$

Using

$$Y_{1,j} = e^{\bar{A}_j^\top T} X_j e^{\bar{A}_j T}, \quad Y_{2,j} = \int_0^T e^{\bar{A}_j^\top \tau} C^\top C e^{\bar{A}_j \tau} d\tau,$$

the Lyapunov–Metzler matrices

$$\mathcal{M}_i = \bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j \ne i} \pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C$$

are computed. Numerical evaluation gives

$$\lambda_{\max}(\mathcal{M}_1) = -35.94, \quad \lambda_{\max}(\mathcal{M}_2) = -68.32.$$

Hence, $\mathcal{M}_1 < 0$ and $\mathcal{M}_2 < 0$, verifying the feasibility of the Lyapunov–Metzler inequalities.

D. Switching Law Implementation

Let $\sigma(t_k) = i$ be the active topology at the switching instant t_k. The topology is first held for the dwell-time interval $[t_k, t_k + T]$ with $T = 0.2$. For $t > t_k + T$, a switch is triggered if there exists $j \ne i$ such that

$$\xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t) < 0.$$

Accordingly, the next switching instant and the next active topology are given by

$$t_{k+1} = \inf_{t > t_k + T}\left\{t : \exists j \ne i,\ \xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t) < 0\right\},$$

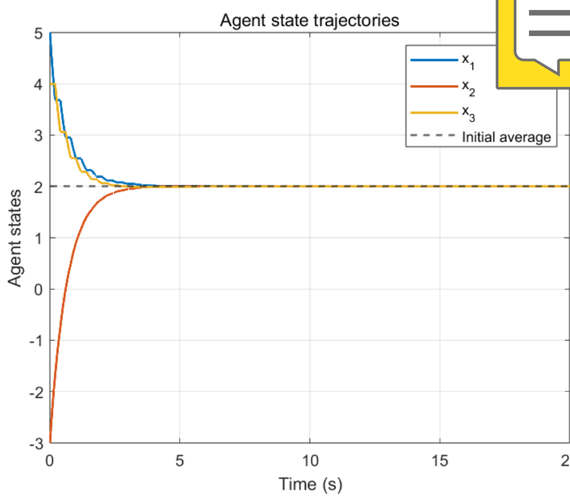

Fig. 1. Agent state trajectories under the proposed state-dependent dwell-time switching law.

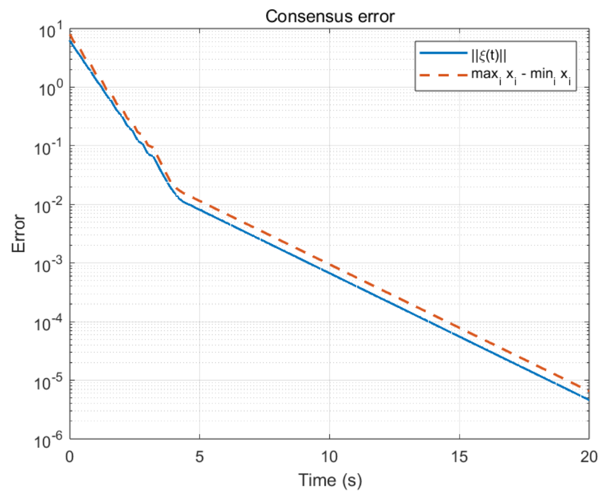

Fig. 2. Consensus error evolution.

and

$$\sigma(t_{k+1}) = \arg\min_{j \neq i} \xi^\top(t_{k+1})\,(Y_{1,j} + Y_{2,j})\,\xi(t_{k+1}).$$

Thus, the switching law selects the topology with the minimum predicted cost while satisfying the dwell-time constraint.

E. Initial Conditions and Simulation Parameters

The initial state is chosen as

$$x(0) = \begin{bmatrix} 5 & -3 & 4 \end{bmatrix}^\top.$$

The corresponding initial average is $\bar{x}(0) = 2$. Since the candidate topologies are weight-balanced, the average state is preserved, and the expected consensus value is 2. The simulation is performed over $t_f = 20\,\mathrm{s}$ with the integration step $\Delta t = 0.001\,\mathrm{s}$. The state trajectories, the disagreement error norm $\|\xi(t)\|$, and the switching signal $\sigma(t)$ are recorded.

F. Simulation Results

The simulation results show that the three agent states converge to the initial average $\bar{x}(0) = 2$. The state trajectories are shown in Fig. 1. It can be observed that all agent states approach the same value, although the system switches between two individually non-strongly-connected topologies.

At $t = 20\,\mathrm{s}$, the terminal state is approximately

$$x(20) \approx \begin{bmatrix} 2.00000322 & 2.00000007 & 1.99999671 \end{bmatrix}^\top.$$

Moreover, the terminal disagreement error norm and the maximum state difference are

$$\|\xi(20)\| \approx 4.60\times10^{-6}, \quad \max_i x_i(20) - \min_i x_i(20) \approx 6.51\times10^{-6}.$$

The evolution of the consensus error is shown in Fig. 2, which confirms that both $\|\xi(t)\|$ and $\max_i x_i(t) - \min_i x_i(t)$ converge to nearly zero.

The switching signal is shown in Fig. 3. It can be seen that the system switches between the two candidate topologies, and the time interval between any two consecutive switches is no less than the prescribed dwell time $T = 0.2$. Therefore, the proposed switching law achieves consensus while satisfying the dwell-time constraint.

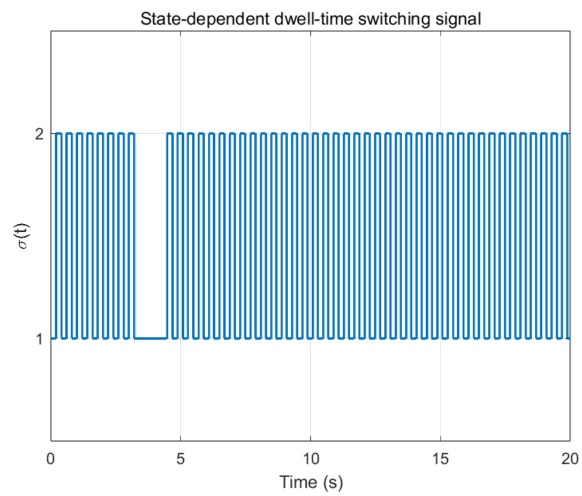

Fig. 3. State-dependent dwell-time switching signal.

VI. Conclusion

This paper first transformed the multi-agent consensus problem into the stability problem of a reduced-order error system. Since each candidate topology is not strongly connected, no individual subsystem can guarantee error convergence. The strong strong connectivity of the union graph was then utilized to exclude common non-consensus zero directions, demonstrating that the system has no structural obstacles to consensus. On this basis, Lyapunov–Metzler inequalities were introduced to characterize the switching coupling relationships among different topologies. Finally, a state-dependent dwell-time switching law was designed to enable the system to switch appropriately among multiple non-strongly-connected directed topologies, thereby achieving global consensus.

The key feature of the proposed method is that it does not require any candidate directed topology to be connected, nor does it rely on a Hurwitz average system as a necessary condition. Instead, it leverages the joint strong-connectivity structure of multiple disconnected directed topologies and the Lyapunov–Metzler dwell-time switching law to achieve global consensus. This provides a new analytical perspective for the consensus control of multi-agent systems with communication constraints, intermittently available topologies, or block-wise varying network connections.

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